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ESTIMATION OF THE RELIABILITY OF RATINGS*

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A procedure for estimating the reliability of sets of ratings, test scores, or other measures is described and illustrated. This procedure, based upon analysis of variance, may be applied both in the special case where a complete set of ratings from each of k sources is available for each of n subjects, and in the general case where $k_1, k_2, \cdots, k_n$ ratings are available for each of the n subjects. It may be used to obtain either a unique estimate or a confidence interval for the reliability of either the component ratings or their averages. The relations of this procedure to others intended to serve the same purpose are considered algebraically and illustrated numerically.

The Problem

The process of estimating test reliability by correlating two sets of scores is well known. The two sets of scores are usually obtained from two equivalent forms, split halves, or two administrations of the test. But when one is dealing with measures other than test scores, such as performance ratings, it frequently happens that more than two parallel sets are available. For example, in one study which concerned us recently, nineteen English instructors graded each of five themes. We desired a measure of their agreement with each other, both before they had received special training in theme rating and again after that training. A complete table of ratings was available in this study, since each instructor rated each of the themes.

In other similar studies, however, the available sets of ratings are sometimes incomplete. An example of this is provided by a study in which eight physics professors rated the research potentialities of twenty-two graduate students. We wished to measure the agreement among the raters in order to establish the reliability of the average ratings as a criterion for validating a selection test. Each professor was asked to rate only those students whose work he knew well. Hence the table of ratings in this study was incomplete.

Problems similar to those just mentioned arise frequently in educational and psychological research. Several formulas have been pro-

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posed to deal with them, but there has been no general agreement on a best method. Peters and Van Voorhis (8) present a formula for average intercorrelation (No. 118) which is appropriate in certain situations. While this formula is derived on the basis of a complete table of ratings, further derivation leads to another formula (No. 119) which may be used where ratings are incomplete. Clark (1) has reported a study in which such a formula was applied to data which did not provide a complete table of ratings.

Fisher's work on intraclass correlation (3) has led to a formula based upon the analysis of variance. This formula is presented in convenient form by Snedecor (9). It is applicable to either complete or incomplete sets of ratings. Horst (5) developed a generalized formula for the reliability of measures which is also applicable to either complete or incomplete sets of ratings. Horst's formula yields the reliability of *average* ratings. However, the Spearman-Brown transformation may be used to obtain the reliability of individual ratings.

At this point a question is likely to arise. Is it better to estimate the reliability of individual ratings or the reliability of average ratings? If decisions are based upon average ratings, it of course follows that the reliability with which one should be concerned is the reliability of those averages. However, if the raters ordinarily work individually, and if multiple scores for the same theme or student are only available in experimental situations, then the reliability of individual ratings is the appropriate measure. Since the reliability of average ratings is determined completely by the reliability of the component ratings, and by the number of components, it is always possible to determine the reliability of individual ratings, or of averages, no matter which value a formula gives initially. Formulas using both approaches will be presented in the following section.

A somewhat different approach to the problem of rater agreement has been suggested by Gulliksen (4). This approach, based on the Wilks-Votaw tests for compound symmetry, does not yield a quantitative estimate of the *degree of agreement* between ratings, but provides instead a statistical criterion upon which to base a categorical statement that the raters do or do not agree. If the distributions of scores from different sources are similar enough so that one can not reject at the five per cent level of confidence the hypothesis that the sets of ratings are random samples from the same population, Gulliksen recommends that the sources of ratings be regarded as parallel (or in agreement). If this hypothesis can be rejected at the one per cent level of confidence, he recommends that the sources

be regarded as not parallel. Gulliksen's suggestion has another application in the study of rater agreement. If one has *a priori* reason to believe that different "schools of thought" may exist among the raters, it is possible to use the Wilks-Votaw tests to check this hypothesis. One should not, however, test the hypothesis on the same set of data that suggested it.

The Intraclass Correlation Formula

When the formulas of Peters, Snedecor, and Horst for estimating the reliability of ratings were applied to the same sets of data, they yielded some inconsistent coefficients. An analysis of the sources of these inconsistencies has led to the conclusion that the formula for intraclass correlation is the most convenient and generally useful. The derivation of this formula is outlined here, since it has not been widely used in studies of educational and psychological ratings, and since few textbooks on measurement contain any discussion of it.

Suppose we have a sample of k estimates of a trait in each of a sample of n persons. Each estimate may be considered to consist of a true component and an error. The true component is constant in all k estimates for any one person, but varies from person to person. Let A represent the variance of these true components in the population of persons from which we have sampled.

The error component varies from estimate to estimate for the same person, but this variance is assumed to be substantially the same in all sets of ratings for the various persons. Let B represent the variance of these errors in the population of estimates. The total observed variance of the estimates is thus $A + B$.

The reliability of the estimates is defined as that portion of the observed variance which is true variance, or

$$r = \frac{A}{A + B}. \tag{1}$$

Suppose we have analyzed the variance of the foregoing sample of estimates to obtain a mean square for error (M) and a mean square for persons (M_p). The mean square for error is a direct estimate of B, the variance of the population of errors of estimate, or

$$M = B. \tag{2}$$

The mean square for persons, however, is not a direct estimate of A. Rather, it represents k times the variance of the means of the estimates for each of the n persons. The variance of these n means is not

attributable to A alone, but also includes an error component attributable to B. Each mean consists of a true component, drawn from a population with variance A, and an error component, which is the mean of k errors drawn from a population with variance B. Hence the variance of the means is $A + B/k$. The mean square for persons is k times the variance of the means. Hence

$$M_{\bar{r}} = kA + B. \tag{3}$$

Solving equations (2) and (3) for A and B, and substituting these values in formula (1), we obtain the formula given by Snedecor,

$$r_1 = \frac{M_{\bar{r}} - M}{M_{\bar{r}} + (k-1)M}. \tag{4}$$

This is the formula for the reliability of individual ratings. Cureton (2) suggests the following parallel derivation of a formula for the reliability of average ratings.

For the average scores, the variance ratio analogous to (1) is

$$r_k = \frac{A}{A + \overline{B}}, \tag{a}$$

where $\overline{B}$ is the error variance of the person means. The estimate of A is still given by (3) and (2); i.e.,

$$A = \frac{M_{\bar{r}} - M}{k}, \tag{b}$$

and the estimate of $\overline{B}$ is given by the usual formula for the error variance of a mean,

$$\overline{B} = \frac{M}{k}. \tag{c}$$

Substituting from (b) and (c) in (a), we obtain at once,

$$r_k = \frac{M_{\bar{r}} - M}{M_{\bar{r}}}. \tag{d}$$

It is worth noting that formula (d) may also be derived by applying the Spearman-Brown formula to formula (4) given above.

TABLE 1
Analysis of Ratings for Problem 1—Complete Sets

	Rater 1	Rater 2	Sum	Sum²
Pupil 1	3	1	4	16
Pupil 2	1	3	4	16
Pupil 3	5	4	9	81
Pupil 4	4	5	9	81
Sum	13	13	26	194
Sum²	169	169	338	
Sum of squared ratings			=	102
Product of sum and mean	$26 \times \frac{26}{8}$		=	84.5
Sum of squares				
For raters	$\frac{338}{4} - 84.5$		=	0.0
For pupils	$\frac{194}{2} - 84.5$		=	12.5
For total	$102 - 84.5$		=	17.5
For error	$17.5 - 12.5 - 0.0$		=	5.0
Mean square				
For pupils	$12.5 \div 3$		=	4.1667
For error	$5.0 \div 3$		=	1.6667
Reliability of ratings	$\frac{4.1667 - 1.6667}{4.1667 + (2-1)\ 1.6667}$		=	.4286
Reliability of average ratings	$\frac{4.1667 - 1.6667}{4.1667}$		=	.6000

Illustrations

Snedecor's formula is applied to a simple problem involving a complete table of ratings in Table 1. Those not familiar with analysis of variance may refer to Table 4 for the formulas used. In this analysis, three components, attributable to pupils, raters, and error, may be separated. Thus it is possible, if desired, to remove the "between-raters" variance from the error term. This overcomes the chief objection of Peters and Van Voorhis to intraclass correlation coefficients, which is that such coefficients are seriously distorted by differences between raters in general level of rating. In Table 1, the "between-raters" variance is zero, so retention or removal gives the same result.

Whether or not it is desirable to remove "between-raters" variance in estimating the reliability of ratings depends upon the way

in which the ratings are ultimately used in grading, classification, or selection. In any case where differences from rater to rater in general level of rating do not lead to corresponding differences in the ultimate grades, classifications, or selections, the "between-raters" variance should be removed from the error term. Specifically, the "between-raters" variance *should be removed* where the final ratings on which decisions are based consist of averages of complete sets of ratings from all observers, or ratings which have been equated from rater to rater such as ranks, Z-scores, etc. Likewise, if comparisons are never made practically, but only experimentally, between ratings of pupils by different raters, the "between-raters" variance should be removed. But if decisions are made in practice by comparing single "raw" scores assigned to different pupils by different raters, or by comparing averages which come from different groups of raters, then the "between-raters" variance should be included as part of the error terms.

TABLE 2

Analysis of Ratings for Problem 5—Incomplete Sets

	Ratings	k	Sum
Pupil 1	8 6 4 4 3	5	25
Pupil 2	6 9 9 4 9 6 5 10 8	9	66
Pupil 3	4 9 10	3	23
	Sums	17	114
Sum of squared ratings			858
Sum of products (pupil sum times pupil mean)			785.3333
Product of sum and mean			764.4706
Sum of squares			
For total	858 — 764.4706	=	93.5294
For pupils	785.3333 — 764.4706	=	20.8627
For error	93.5294 — 20.8627	=	72.6667
Mean square			
For pupils	20.8627 ÷ 2	=	10.4314
For error	72.6667 ÷ 14	=	5.1905
Average value of k			5.1176
Reliability	$\frac{10.4314 - 5.1905}{10.4314 + (4.1176)(5.1905)}$	=	.1648

Table 2 illustrates application of this formula to a simple problem in which the table of ratings is incomplete and the sources of ratings are not identified. In this case only two components of the variance, attributable to pupils and error, are separated. Thus any difference in general level of ratings between the various raters is automatically included in the error term.

The application of this reliability formula in Table 2 presents a special problem. The formula requires a value of k, the number of ratings of each person. But in Table 2 this number is not the same for each person. Snedecor (9, p. 234) suggests the following formula for an average k:

$$k_0 = \frac{1}{n-1}\left[\sum k - \frac{\sum k^2}{\sum k}\right]. \qquad (5)$$

The average k thus obtained is approximately the harmonic mean of the k's for each pupil.

Estimates of Precision

When the reliability of ratings is estimated on the basis of a sample of products or raters or both, a description of the precision of the reliability estimate obtained is useful in judging the adequacy of the sample or the confidence which can be placed in the obtained estimate. The intraclass formula lends itself readily to such a description. The method used here was suggested by Jackson and Ferguson's description of confidence intervals for their sensitivity coefficient (6).

The first step in obtaining this estimate is to express the variance between products, and the error variance as a single ratio, here designated by F_s. In equation form

$$F_s = \frac{M_{\bar{x}}}{M}. \qquad (6)$$

Using this ratio it is possible to transform the intraclass formula as follows:

$$r = \frac{F_s - 1}{(F_s - 1) + k}. \qquad (7)$$

Now if the sample values of $M_{\bar{x}}$ and M are considered as having been drawn from separate populations, it is obvious that the ratio (F_p) for the populations might be either greater or less than the ratio observed in the sample. How much greater or less one should believe it to be, at any selected level of confidence, can be read from a Table for F, given the degrees of freedom for the sample values of $M_{\bar{x}}$ and M. To obtain an estimate for the upper limit of the variance ratio between two populations, one must multiply the variance ratio

observed in the samples (F_s) by the maximum ratio expected if the populations had been equal in variance. This value (F_t) may be read in the table, given the number of degrees of freedom in the samples, and the level of confidence desired. In entering the table for F it is important to remember that at the upper limit $M_{\bar{x}}$ is the larger variance, so that the degrees of freedom for $M_{\bar{x}}$ should be located on the marginal headings for the larger variance. Similarly, to obtain the lower limit of (F_p) one must multiply the sample value by the reciprocal of the value given in the table. Remember that at the lower limit, M is the larger variance, so that in this case the degrees of freedom for M should be located on the marginal headings for the larger variance. Substituting these two limiting values for the population variance ratio in formula 6 yields upper and lower limits of the confidence interval for the estimate of reliability obtained from a particular sample.

The application of this procedure to several sample problems is illustrated in Table 3. The data for problems 1 and 3 are taken from

TABLE 3
Confidence Limits (5%) for Reliability Coefficients From Sample Data

Problem	1	3	5
Unique Estimate	.4286	.3429	.1648
k	2.	2.	5.1176
F_s	2.500	2.0435	2.0096
F_t (upper 5% limit)	9.28	9.28	3.74
(lower 5% limit)	9.28	9.28	19.42
F_p (upper 5% limit)	23.125	18.964	7.5159
(lower 5% limit)	.2694	.2202	.1035
r (upper 5% limit)	.92	.90	.56
(lower 5% limit)	—.58	—.64	—.21

Table 5, and the data for problem 5 are taken from Table 2. Because of the smallness of the samples involved, the confidence limits for these reliability coefficients are very wide indeed. Reliability estimates based upon such small samples are little better than blind guesses.

Relationships Among the Formulas

The fact that various formulas have been presented for estimating the reliability of ratings, and that these formulas do not always

yield consistent results when applied to the same set of data was mentioned in the first section of this paper. It is therefore appropriate to consider the relationships among them first analytically and then, by way of illustration, empirically. Consider first the relation between the Pearson product-moment formula and the formula for average intercorrelation given by Peters and Van Voorhis. As usually presented these formulas are

(1) Product moment:

$$r = \frac{\Sigma xy}{N \sigma_x \sigma_y}. \tag{8}$$

(2) Average intercorrelation:

$$r = \frac{(\sigma_s^2/\sigma_i^2) - a}{a^2 - a}. \tag{9}$$

We note that the quantity $\frac{\Sigma xy}{N}$ is the covariance of the scores for each *product* from two *observers*. The symbol ζ will be used to represent this covariance.

It is easy to show (8, p. 196) that the variance of the sums of scores from k observers is

$$\sigma_s^2 = k\,\sigma_i^2 + (k^2 - k)\,\zeta_{ij}. \tag{10}$$

If this value is substituted for σ_s^2 in formula (9) the result is

$$r = \frac{\zeta_{ij}}{\sigma_i^2}. \tag{11}$$

Since the product-moment formula may be written

$$r = \frac{\zeta}{\sqrt{\sigma_x^2\,\sigma_y^2}}, \tag{12}$$

it is clear that when $k = 2$, (the only case in which the product-moment formula is applicable) and when the table of ratings is complete (which must be true for either formula to apply) the numerators are identical. But since σ_i^2, which represents the variance of scores from either observer, is estimated by taking the *arithmetic mean* of the variances of scores from each observer, σ_i^2 will not equal the geometric mean of these variances, $\sqrt{\sigma_x^2\,\sigma_y^2}$ except in the special case where $\sigma_x^2 = \sigma_y^2$. Hence, differences in coefficients obtained from these two formulas are attributable to the difference in methods of calcu-

lating the average variance within observers.

It is interesting to note in passing that ζ provides a direct estimate of A, the variance of the population of true scores. Recalling the definitions of A and B given earlier in this paper, one may see that

$$\sigma_s^2 = k^2 A + k B \tag{13}$$

TABLE 4
Notation for Analysis of Ratings
(Complete Arrays)

A. Data

	k observers					Sums
n products	0_1	0_2	0_3	0_j	0_k	
P_1	x_{11}	x_{12}	x_{13}	x_{1j}	x_{1k}	P_1
P_2	x_{21}	x_{22}	x_{23}	x_{2j}	x_{2k}	P_2
P_3	x_{31}	x_{32}	x_{33}	x_{3j}	x_{3k}	P_3
P_i	x_{i1}	x_{i2}	x_{i3}	x_{ij}	x_{ik}	P_i
P_n	x_{n1}	x_{n2}	x_{n3}	x_{nj}	x_{nk}	P_n
Sums	0_1	0_2	0_3	0_j	0_k	T

B. Analysis of Variance

Source	Degrees of Freedom	Sums of Squares	Mean Square
Products	$n-1$	$S_p = \frac{\Sigma P^2}{k} - \frac{T^2}{nk}$	$M_p = \frac{S_p}{n-1}$
Observers	$k-1$	$S_0 = \frac{\Sigma 0^2}{n} - \frac{T^2}{nk}$	$M_0 = \frac{S_0}{k-1}$
Error	$(n-1)(k-1)$	$S_e = S_t - S_p - S_0$	$M_e = \frac{S_e}{(n-1)(k-1)}$
Total	$nk-1$	$S_t = \Sigma x^2 - \frac{T^2}{nk}$	

and

$$\sigma_t^2 = A + B . \tag{14}$$

Substituting these in (9) and simplifying leads to the expression

$$A = \zeta . \tag{15}$$

Consider next the relation between the formulas for average intercorrelation (9) and intraclass correlation (4). The notation used in this and the following comparison is presented in Table 4. Since

$$\sigma_s^2 = \frac{\Sigma P^2}{n} - \left(\frac{T}{n}\right)^2$$

or

$$n\,\sigma_s^2 = \Sigma P^2 - \frac{T^2}{n},$$

and since

$$S_p = \frac{\Sigma P^2}{k} - \frac{T^2}{nk}$$

or

$$k\,S_p = \Sigma P^2 - \frac{T^2}{n},$$

it follows that

$$\sigma_s^2 = \frac{k\,S_p}{n}. \tag{16}$$

Further, since

$$\sigma_i^2 = \frac{\Sigma\Sigma x^2}{nk} - \frac{\Sigma 0^2}{n^2k}$$

or

$$n\,k\,\sigma_i^2 = \underline{\Sigma\Sigma}\, x^2 - \frac{\underline{\Sigma 0^2}}{n},$$

and since

$$S_e + S_p = S_t - S_0 = \Sigma\Sigma x^2 - \frac{\underline{\Sigma 0^2}}{n},$$

then

$$\underline{\sigma_i^2} = \frac{S_e + S_p}{nk}. \tag{17}$$

Substituting the foregoing values of σ_s^2 and σ_i^2 in formula (9) gives, upon simplification,

$$r = \frac{S_p - \dfrac{S_e}{k-1}}{S_p + S_e}. \tag{18}$$

But

$$S_p = (n-1)M_p$$

and

$$S_e = (n-1)(k-1)M_e.$$

Substituting these values in formula (18) and simplifying gives

$$r = \frac{M_p - M_e}{M_p + (k-1)M_e} = \frac{M_{\bar{x}} - M}{M_{\bar{x}} + (k-1)M}.$$

In terms of these derivations, the average intercorrelation appears identical with the intraclass correlation formula. This will be true, however, only if the "between-raters" variance is not included as part of the error variance in either formula, or if it *is* included in both. As was previously pointed out, some situations require inclusion of the "between-raters" variance and others do not. The user of the intraclass formula finds it convenient to choose either procedure. The user of the average intercorrelation formula may also exercise an option in this matter. He may include the "between-raters" variance by calculating the average "within-raters" variance (σ_i^2) about the general mean for all raters. Or, he may exclude it by calculating that variance about the mean for each rater separately, as was indicated in the foregoing derivation.

It is also worth noting that the formulas give identical coefficients in spite of the fact that the average intercorrelation formula uses sample statistics whereas the intraclass formula uses estimates of population parameters. Since the reliability coefficient is basically a ratio of true and observed variances, application of the sample correction to both variances of the ratio does not change its value.

Consider finally the relation between the intraclass correlation formula and generalized formula for the reliability of averages. The latter, as given by Horst, is

$$r_{av} = 1 - \frac{\dfrac{\Sigma \dfrac{\sigma_i^2}{n_i - 1}}{N}}{\sigma_M^2}. \tag{19}$$

In the case where a complete table of ratings is available so that $n_1 = n_2 = n_3 = \cdots\cdots = n_i = k$, the formula may be written (substituting n for Horst's N)

$$r_{av} = 1 - \frac{\sum \sigma_i^2}{n(k-1)\sigma_M^2}. \tag{20}$$

The formula relating the reliability of an average of k equally weighted scores to the reliability of the scores themselves is

$$r_{av} = \frac{kr}{1 + (k-1)r}. \tag{21}$$

If the right hand members of equations (20) and (21) are equated and the equation solved for r, we obtain

$$r = \frac{\sigma_M^2 - \dfrac{\sum \sigma_i^2}{n(k-1)}}{\sigma_M^2 + \dfrac{\sum \sigma_i^2}{n}}. \tag{22}$$

Now since

$$\sigma_M^2 = \frac{\sum\left(\frac{P}{k}\right)^2}{n} - \left(\frac{T}{nk}\right)^2$$

or

$$nk^2\,\sigma_M^2 = \sum P^2 - \frac{T^2}{n},$$

and since, as was shown previously,

$$k\,S_p = \sum P^2 - \frac{T^2}{n},$$

then

$$n\,k^2\,\sigma_M^2 = k\,S_p$$

or

$$\sigma_M^2 = \frac{1}{n\,k}\,S_p.$$

Since, further,

$$\sum \sigma_i^2 = \frac{\sum\sum x^2}{k} - \sum\left(\frac{P}{k}\right)^2$$

or

$$k\sum \sigma_i^2 = \sum\sum x^2 - \sum \frac{P^2}{k},$$

and since

$$S_0 + S_e = S_t - S_p = \sum\sum x^2 - \frac{\sum P^2}{k},$$

then

$$k \sum \sigma_i^2 = S_0 + S_e$$

or

$$\sum \sigma_i^2 = \frac{S_0 + S_e}{k}.$$

Substituting the above values for σ_M^2 and $\sum \sigma_i^2$ in formula (22), we find

$$r = \frac{S_p - \dfrac{(S_0 + S_e)}{k-1}}{S_p + (S_0 + S_e)}. \tag{23}$$

Comparing this formula with formula (18) we observe that the sole difference is that the "error term" of formula (23) includes the "between-raters" sum of squares, whereas the formula (18) does not.

Hence, whenever there is a difference between the means of the ratings from various raters, the generalized reliability formula will give a lower value for the reliability of the ratings than is given by the intraclass formula as here calculated. The circumstances of each problem determine whether the "between-raters" variance should be included as part of the error term. As pointed out previously, the user of the intraclass formula may easily include or exclude "between-raters" variance as part of the error term. The user of the generalized reliability formula can not conveniently exclude "between-raters" variance even where it appears desirable to do so. In particular, the generalized reliability formula should not be used in estimating reliability of averages to which each rater has contributed by rating each product. Attention was called earlier to the fact that "between-raters" variance does not belong in the error term in this situation.

In this connection it should be mentioned that "between-raters" variance is always removed in the process of calculating the product-moment formula. Hence, as Gulliksen has pointed out in a private communication, the product-moment formula should not be used in cases where the "between-raters" variance is properly part of the error.

Both the intraclass formula and the generalized reliability formula are applicable to situations where the table of ratings is incomplete, that is, where k varies from product to product. But here again

a computational difference appears which results in different reliability coefficients. With the intraclass formula the mean square for error based on two component analysis is

$$Me = \frac{1}{N-n}\left[\Sigma\Sigma x^2 - \frac{\Sigma P^2}{k}\right], \qquad (24)$$

so that each score contributes equally to the estimate of error variance. With the generalized reliability formula, however, the error variance for each product is figured separately and the separate error variances are then averaged, as indicated by the numerator of the fraction in formula (19). A "within-product" variance based on many scores is given the same weight as one based on few scores. Thus, a score in a small group is weighted more heavily than one in a large group.

A similar difference is observed in calculation of the variance of the means for products. In analysis of variance each product mean is weighted in proportion to the number of scores on which it is based. But in the generalized reliability formula each product mean is given equal weight. Since unweighted averages are seldom the same as weighted averages from the same data, the reliability estimates obtained when in the two formulas are applied to incomplete tables of scores are usually different.

Table 5 presents four sample problems, the data of which illustrate various combinations of equal or unequal means and variances. Below the data are given the reliability coefficients obtained by ap-

TABLE 5
Rater Reliability by Various Formulas for Problems Illustrating Various Conditions

Problem	1	2	3	4
Condition				
Means	Equal	Unequal	Equal	Unequal
Variances	Equal	Equal	Unequal	Unequal
Scores (Test 1 and Test 2)				
Pupil 1	3 1	3 3	6.25 2	3 2
Pupil 2	1 3	1 5	4.25 6	1 6
Pupil 3	5 4	5 6	8.25 8	5 8
Pupil 4	4 5	4 7	7.25 10	4 10
Reliability				
Product-Moment	.4286	.4286	.4286	.4286
Intraclass	.4286	.4286	.3429	.3429
Average Intercorrelation	.4286	.4286	.3429	.3429
Generalized Reliability	.4286	—.0196	.3429	—.0944

plying each of the four formulas which have just been discussed. The values obtained confirm the analytical findings.

In Problem 1, with equal means and variances, all four formulas yield identical coefficients of reliability. In Problem 2, where only the means of the two sets of scores differ, the generalized reliability formula gives a much lower coefficient than the other three. This is due to the previously noted fact that systematic differences between the sets of scores are included as part of the error term in the generalized reliability formula, whereas these differences are removed in the other three formulas.

In Problem 3, where only the variances of the two sets of scores differ, all three of the special formulas yield coefficients which disagree with that from the product-moment formula, but which agree perfectly with each other. This is due to the fact that the product-moment formula uses a geometric mean of the variances of the scores from each observer, whereas all of the other formulas are based upon arithmetic means of these variances. In Problem 4, both the means and the variances of the two sets of scores differ, with the result that the intraclass coefficient and the average intercorrelation coefficient are somewhat lower, and the generalized reliability coefficient is very much lower, than the product-moment coefficient.

If the generalized reliability formula is applied to the data of Table 2 it yields a reliability coefficient of —.0582 compared with .1648 given by the intraclass formula. Here the discrepancy occurs because the generalized reliability formula uses a simple mean of the error variances within each person, and a simple variance of the person means, whereas the analysis of variance formula uses a weighted mean and a weighted variance.

The formula used by Clark was

$$r = \frac{(a\,\sigma_{av}^2/\sigma_i^2) - 1}{a - 1}. \tag{25}$$

The σ_{av}^2 is the variance of the averages for each product, and the σ_i^2 is, in this case, the total variance of all ratings. When this formula is applied to the data of Table 2 it yields a reliability coefficient of .2060 instead of the .1648 given by the intraclass formula. Here the discrepancy is partly due to the use in Clark's formula of a simple variance of the pupil means, and partly due to the estimation of the variance of the population of ratings from three sets of related ratings, which are treated as if they constituted a single random sample of the population of all ratings. Fisher (3, 225) has called attention to the error

introduced by the second of these procedures. These two differences in procedure no doubt cause much more apparent error in the small illustrative sample presented in Table 2 than they did in the more extensive data upon which Clark based his report. However the possibility of such errors when formula (25) is applied to incomplete sets of ratings should be recognized.

Conclusions

In view of the foregoing findings, there are three reasons for preferring the intraclass formula to the average intercorrelation or generalized reliability formulas. First, the intraclass formula permits the investigator to choose whether to include or exclude "between-raters" variance as part of the error variance, in terms of the circumstances of the particular problem. Second, a convenient means for estimating the precision of the reliability coefficients is available to the user of the intraclass formula. Third, the intraclass formula uses the familiar statistics and routine computational procedures of analysis of variance.

In the case of incomplete sets of ratings, the intraclass formula also has advantages over the generalized reliability formula and the variant of the average intercorrelation formula which Clark used in this situation. First, in determining the variances used in the intraclass formula, each observation is weighted equally, whereas in both the other formulas it is groups of unequal numbers of observations that are weighted equally. Second, Clark's formula involves a biased estimate of the population variance. Third, the advantages of the analysis of variance approach in estimating the precision of the reliability coefficient, and in computation, also apply in the case of unequal sets of ratings.

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